

SOFTWARE REQUIREMENTS SPECIFICATION

Nutrition & Fitness Calculator Hub

NutriCalc Pro — Static Site + USDA API Integration

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Author	[Your Name], [Your Credentials]
Target Stack	HTML5 / CSS3 / Vanilla JS + Eleventy (11ty)
Hosting	Cloudflare Pages (free tier)
Backend	None (client-side only, except Recipe Analyzer)
API Dependency	USDA FoodData Central (free, no auth required)

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the complete technical requirements, architecture decisions, calculator logic, SEO strategy, API integrations, and deployment pipeline for NutriCalc Pro — a credential-backed, static nutrition and fitness calculator hub designed to rank for long-tail search terms and serve genuine user value through scientifically-cited, author-attributed tools.

This document is the single source of truth for all development work. It is written to be executed directly by a developer (solo or small team) without ambiguity.

1.2 Scope

NutriCalc Pro comprises:

- 9+ client-side calculators built in vanilla JavaScript with zero runtime dependencies
- 1 server-side-adjacent recipe analyzer powered by the USDA FoodData Central REST API (client calls made directly from the browser — no custom backend)
- Static site generation via Eleventy (11ty) for fast builds, templating, and SEO-friendly HTML output
- Deployment on Cloudflare Pages with edge caching, custom domain, and free SSL
- A structured content layer: author bio, per-page supporting articles, references, schema markup
- Condition-specific tools (diabetic carb counting, kidney-friendly macro limits) that establish medical credibility in extremely low-competition SERPs

1.3 Definitions, Acronyms & Abbreviations

Term	Definition
TDEE	Total Daily Energy Expenditure — total calories burned per day including activity
BMR	Basal Metabolic Rate — calories burned at complete rest
Mifflin-St Jeor	1990 BMR formula; most accurate for general population ($\pm 10\%$)
Katch-McArdle	BMR formula using lean body mass; superior when body fat % is known
Navy Method	U.S. Navy body fat % estimation using circumference measurements
USDA FDC	USDA FoodData Central — free nutrition database REST API
SSG	Static Site Generator — builds HTML at deploy time, not request time

Term	Definition
11ty / Eleventy	Zero-config JavaScript SSG; outputs pure HTML with no client-side framework overhead
LCP	Largest Contentful Paint — Core Web Vital; target < 2.5s
CLS	Cumulative Layout Shift — Core Web Vital; target < 0.1
SERP	Search Engine Results Page
Long-tail	Search queries with 3+ words, lower volume, much lower competition
E-E-A-T	Experience, Expertise, Authoritativeness, Trustworthiness — Google quality signals
JSON-LD	JSON Linked Data — format for embedding structured schema markup in HTML
CSP	Content Security Policy — HTTP header controlling resource loading
DXA	Document eXtended Attribute unit (1440 DXA = 1 inch) — used only in this .docx

1.4 Intended Audience

- Primary: The developer(s) building NutriCalc Pro
- Secondary: The content author (you) for article and bio requirements
- Tertiary: Future collaborators or contractors onboarding to the project

1.5 Assumptions & Constraints

- All calculator math runs client-side; no user data is sent to any server
- USDA FoodData Central API is free and does not require authentication for public data endpoints
- Cloudflare Pages free tier supports unlimited static asset bandwidth
- No database, no user accounts, no cookies (except optional analytics)
- Author credentials (degree, certifications) are real and can be cited truthfully
- All content is original — no AI-generated boilerplate published as-is

2. Overall System Description

2.1 Product Perspective

NutriCalc Pro is a standalone static website. It has no dependencies on any external CMS, database, or authentication service. The architecture is deliberately simple: Eleventy transforms Nunjucks/Markdown templates into flat HTML files at build time. These files are served directly from Cloudflare's edge network with no origin server, achieving sub-100ms TTFB globally.

The sole external runtime dependency is the USDA FoodData Central API, called directly from the user's browser in the Recipe/Ingredient Analyzer tool only. All other calculators execute purely in the user's browser with no network calls.

2.2 Product Functions (High Level)

Module ID	Calculator Name	Type	Target Keyword (Primary)
CALC-01	TDEE / BMR Calculator	Client-side JS	TDEE calculator Mifflin St Jeor
CALC-02	Macro Calculator	Client-side JS	macro calculator for cutting
CALC-03	Body Fat % Calculator (Navy)	Client-side JS	Navy body fat calculator
CALC-04	Protein Intake Calculator	Client-side JS	how much protein per day calculator
CALC-05	Fiber Intake Calculator	Client-side JS	daily fiber intake calculator by age
CALC-06	Water Intake Calculator	Client-side JS	water intake calculator by weight kg
CALC-07	Pregnancy Calorie Calculator	Client-side JS	calorie calculator during pregnancy trimester
CALC-08	Recipe Nutrition Analyzer	Client-side JS + USDA API	recipe nutrition calculator free paste ingredients
CALC-09	Diabetic Carb Counter	Client-side JS	carb counting calculator for diabetics
CALC-10	Kidney Diet Calculator	Client-side JS	potassium phosphorus calculator kidney disease

2.3 User Characteristics

- General public seeking free, trustworthy nutrition calculators — no technical knowledge assumed
- People managing chronic conditions (diabetes, CKD) who need specialized tools unavailable on generic sites
- Pregnant and breastfeeding individuals seeking trimester-specific guidance
- Fitness enthusiasts tracking macros for cut / maintain / bulk phases
- Healthcare students or practitioners doing quick reference lookups

i NOTE

All users are anonymous. No PII is collected. No form submissions leave the user's device (except recipe analyzer queries to USDA — which contain only food names, not user identity).

2.4 Operating Environment

Environment	Requirement
Browser Support	Chrome 90+, Firefox 88+, Safari 14+, Edge 90+ (covers 98%+ of traffic)
Mobile	Fully responsive; calculators usable on 375px viewport width (iPhone SE)
JavaScript	Required; graceful no-JS message shown if disabled
Internet Connection	Required only for Recipe Analyzer (USDA API calls). All other calculators offline-capable.
Screen Reader	ARIA labels on all inputs and results. WCAG 2.1 AA compliance required.
Hosting	Cloudflare Pages — global CDN, 0ms cold starts (static files), free SSL

3. Technology Stack & Architecture

3.1 Stack Decision Matrix

Layer	Choice	Alternatives Rejected	Rationale
SSG	Eleventy (11ty)	Next.js, Gatsby, Hugo	Zero JS shipped to client by default. Nunjucks templating is simple. No React overhead. Perfect PageSpeed scores trivially achievable.
Templating	Nunjucks (.njk)	Liquid, Handlebars	Full JavaScript in templates. Mature. Ships with 11ty.
CSS	Custom CSS (no framework)	Tailwind, Bootstrap	No unused CSS. Full control. No purge step. Single 15KB stylesheet.
JS (Calculators)	Vanilla ES6+ modules	React, Vue, Alpine.js	Zero bundle size. No hydration. Calculators are pure functions with simple DOM updates.
Hosting	Cloudflare Pages	Netlify, Vercel, GitHub Pages	Free unlimited bandwidth. Workers KV available if needed. Best CDN globally. Free analytics.
API	USDA FoodData Central	Nutritionix (paid), Edamam (limited free)	Completely free. No API key needed for public endpoints. 300k+ foods. USDA = authoritative citation.
Analytics	Cloudflare Web Analytics	Google Analytics	Privacy-first. No cookies. No consent banner needed. Free.
CI/CD	GitHub Actions → Cloudflare Pages	Manual deploy	Push to main → auto-build → deploy in ~30 seconds.
Search	Pagefind (static search)	Algolia, Elasticsearch	Runs entirely in browser. Index built at SSG time. Zero cost.

3.2 Project Directory Structure

The following is the canonical file tree for the project:

```
nutri-calc-pro/
```

```

├── .github/
│   └── workflows/
│       └── deploy.yml          # GitHub Actions → Cloudflare Pages
├── src/
│   ├── data/
│   │   ├── site.json          # Global site metadata, author info
│   │   └── calculators.json    # Calculator manifest (name, slug, description,
│   │       schema)
│   ├── includes/
│   │   ├── layouts/
│   │   │   ├── base.njk       # HTML shell: head, nav, footer, JSON-LD
│   │   │   └── calculator.njk  # Calculator page layout with article area
│   │   ├── components/
│   │   │   ├── calculator-shell.njk  # Input form + results container
│   │   │   └── author-bio.njk       # Credentials block (appears on every
│   │       page)
│   │   ├── disclaimer.njk       # Legal / not-medical-advice
│   │   └── references.njk       # Numbered citations list
│   ├── calculators/             # One .njk file per calculator page
│   │   ├── tdee.njk
│   │   ├── macros.njk
│   │   ├── body-fat.njk
│   │   ├── protein.njk
│   │   ├── fiber.njk
│   │   ├── water.njk
│   │   ├── pregnancy.njk
│   │   ├── recipe-analyzer.njk
│   │   ├── diabetic-carb.njk
│   │   └── kidney-diet.njk
│   ├── js/
│   │   ├── core/
│   │   │   ├── units.js        # Unit conversion (imperial ↔ metric)
│   │   │   └── validate.js     # Input validation helpers
│   │   ├── calculators/        # One .js module per calculator
│   │   │   ├── tdee.js
│   │   │   ├── macros.js
│   │   │   ├── body-fat.js
│   │   │   ├── protein.js
│   │   │   ├── fiber.js
│   │   │   ├── water.js
│   │   │   ├── pregnancy.js
│   │   │   ├── recipe-analyzer.js
│   │   │   ├── diabetic-carb.js
│   │   │   └── kidney-diet.js
│   │   └── ui/
│   │       ├── form-handler.js # Generic submit → calculate → render
│   │       └── print.js        # Print-friendly results
│   ├── css/
│   │   ├── main.css           # Design tokens, reset, typography
│   │   ├── calculator.css     # Form, results, table styles
│   │   └── print.css          # Print media query overrides
│   ├── articles/              # Supporting markdown articles (1 per calculator)
│   │   ├── tdee.md
│   │   └── ... (one per calculator)
│   ├── index.njk              # Homepage – hub listing all calculators
│   └── .eleventy.js            # Eleventy config

```

```
└─ package.json
└─ _site/
Cloudflare # Build output (gitignored) → deployed to
```

3.3 Eleventy Configuration

The `.eleventy.js` file must configure the following:

- **Input/output dirs:** `src/` → `_site/`
- **Passthrough copy:** `src/js/`, `src/css/`, `src/fonts/`, `src/images/`
- **Markdown:** `markdown-it` with `anchor` plugin for auto-IDs on headings (needed for TOC links)
- **Filters:** `date` formatting, `truncate`, `slugify`
- **Collections:** `calculators` (all `.njk` in `src/calculators/`) for hub page listing and sitemap generation
- **Sitemap:** `eleventy-plugin-sitemap` → `/sitemap.xml` (required for Google Search Console)
- **Minification:** `html-minifier-terser` as transform in production builds

TIP

Do NOT use `eleventy-plugin-bundle` or `Vite/webpack`. Keep the build simple: passthrough JS/CSS, no bundling. Smaller attack surface, faster builds, easier debugging.

4. Calculator Specifications

Each section below defines: inputs, validation rules, mathematical formulas with citations, output fields, and edge cases. All formulas must be implemented exactly as specified. Unit conversion functions are shared from src/js/core/units.js.

4.1 CALC-01: TDEE / BMR Calculator

4.1.1 Inputs

Field ID	Label	Type	Units	Valid Range	Required
age	Age	number	years	15 – 100	Yes
sex	Biological Sex	radio	male / female	—	Yes
weight	Weight	number	kg or lb	30 – 300 kg	Yes
height	Height	number	cm or ft+in	100 – 250 cm	Yes
bodyfat	Body Fat %	number	%	3 – 60	No (enables Katch-McArdle)
activity	Activity Level	select	—	5 options (see below)	Yes
formula	Formula	radio	—	Mifflin / Katch	Yes

4.1.2 Activity Multipliers

Value	Label	Multiplier (Harris-Benedict style)
sedentary	Sedentary (desk job, no exercise)	1.2
light	Lightly Active (1–3 days/week)	1.375
moderate	Moderately Active (3–5 days/week)	1.55
active	Very Active (hard exercise 6–7 days/week)	1.725
extra	Extra Active (physical job + hard training)	1.9

4.1.3 Formulas

Mifflin-St Jeor BMR (primary — cite: Mifflin et al., 1990, AJCN):

```

BMR_male    = (10 × weight_kg) + (6.25 × height_cm) - (5 × age) + 5
BMR_female  = (10 × weight_kg) + (6.25 × height_cm) - (5 × age) - 161

```

Katch-McArdle BMR (cite: Katch & McArdle, 1996 — requires body fat %):

```

LBM = weight_kg × (1 - (bodyfat% / 100))
BMR = 370 + (21.6 × LBM)

```

TDEE:

```

TDEE = BMR × activity_multiplier

```

4.1.4 Outputs

Output Field	Format	Description
BMR	Integer kcal	Calories at rest (rounded to nearest whole number)
TDEE	Integer kcal	Total daily calories at given activity level
Weight Loss (-500 kcal)	Integer kcal	TDEE - 500 (safe ~0.5 kg/week loss)
Weight Gain (+300 kcal)	Integer kcal	TDEE + 300 (lean bulk)
Formula Used	String	Display which formula was applied and why
Macros Preview	g each	Default 40/30/30 C/P/F breakdown of TDEE (link to CALC-02)

💡 TIP

Display BOTH Mifflin and Katch results side-by-side when body fat % is provided. Label clearly. State that Katch-McArdle is more accurate when body fat is known.

4.2 CALC-02: Macro Calculator**4.2.1 Goal Modes**

Mode	Calorie Adjustment	Default Macro Split (C/P/F)	Use Case
Cut (lose fat)	$TDEE \times 0.80$ (20% deficit)	40% / 35% / 25%	Fat loss, preserve muscle
Maintain	$TDEE \times 1.00$	40% / 30% / 30%	Body recomposition
Lean Bulk	$TDEE \times 1.10$ (10% surplus)	45% / 30% / 25%	Muscle gain, minimize fat
Aggressive Bulk	$TDEE \times 1.20$	50% / 25% / 25%	Maximum muscle gain
Custom	User-input kcal	User-input % (must sum to 100)	Advanced users

4.2.2 Formula

```

target_kcal = TDEE × goal_multiplier
carbs_g     = (target_kcal × carb_ratio) / 4
protein_g   = (target_kcal × protein_ratio) / 4
fat_g       = (target_kcal × fat_ratio) / 9

```

Caloric density constants: 4 kcal/g for carbohydrates and protein; 9 kcal/g for fat.

WARNING

Validation: custom macro percentages must sum to exactly 100. Show real-time running total with color feedback (red if $\neq 100$, green if $= 100$).

4.3 CALC-03: Body Fat % Calculator (Navy Method)

4.3.1 Inputs

Field	Units	Valid Range
sex	male / female	—
height	cm	100–250
waist (at navel)	cm	50–200
neck (at larynx)	cm	20–80
hip (widest, female only)	cm	50–200

4.3.2 Formula (U.S. Navy — cite: Hodgdon & Beckett, 1984)

Male: $BF\% = 495 / (1.0324 - 0.19077 \times \log_{10}(\text{waist} - \text{neck}) + 0.15456 \times \log_{10}(\text{height})) - 450$

Female: $BF\% = 495 / (1.29579 - 0.35004 \times \log_{10}(\text{waist} + \text{hip} - \text{neck}) + 0.22100 \times \log_{10}(\text{height})) - 450$

4.3.3 Output: Classification Table

Category	Men (BF%)	Women (BF%)
Essential Fat	2 – 5%	10 – 13%
Athletes	6 – 13%	14 – 20%
Fitness	14 – 17%	21 – 24%
Average	18 – 24%	25 – 31%
Obese	25%+	32%+

4.4 CALC-04: Protein Intake Calculator

Inputs: weight (kg), goal (maintenance / muscle gain / weight loss / endurance), kidney health flag (reduces recommendation if impaired).

4.4.1 Formula (cite: Morton et al., 2018, BJSM; ISSN Position Stand 2017)

Goal	Recommendation	Source
Sedentary / Maintenance	0.8 g/kg	DRI/RDA (Institute of Medicine)
Weight Loss (preserve muscle)	1.2 – 1.6 g/kg	Helms et al., 2014
Muscle Gain / Hypertrophy	1.6 – 2.2 g/kg	Morton et al., 2018
Endurance Athletes	1.4 – 1.7 g/kg	ISSN Position Stand, 2017
Kidney-Impaired (CKD stages 3–5)	0.6 – 0.8 g/kg (no dialysis) / 1.0–1.2 (dialysis)	KDIGO Guidelines 2020

Output: min–max protein range in grams, with visual progress bar against user's TDEE. Show protein as % of total calories.

4.5 CALC-05: Fiber Intake Calculator

Inputs: age, sex, pregnancy status, caloric intake.

4.5.1 Formula (cite: Institute of Medicine, Dietary Reference Intakes, 2005)

Population	Recommendation
Men 19–50	38 g/day
Men 51+	30 g/day
Women 19–50	25 g/day
Women 51+	21 g/day
Pregnancy	28 g/day
Lactation	29 g/day
Alternative (calorie-based)	14 g per 1000 kcal consumed

4.6 CALC-06: Water Intake Calculator

Inputs: weight (kg), activity level, climate (temperate / hot), pregnancy/lactation status.

4.6.1 Formula (cite: IOM, 2005; EFSA, 2010)

```
base_ml = weight_kg * 35
activity_bonus = (activity_minutes / 30) * 350 // 350ml per 30min exercise
climate_bonus = (climate === 'hot') ? base_ml * 0.10 : 0
pregnancy_add = 300 // ml extra
lactation_add = 700 // ml extra
total_ml = base_ml + activity_bonus + climate_bonus + relevant_add
```

Output: total daily water in ml AND cups. Recommend spread across the day: show a simple hourly schedule widget (e.g., drink N ml every 2 hours from 7am to 9pm).

4.7 CALC-07: Pregnancy Calorie Calculator

Competitive advantage: Almost no free tools cover trimester-specific adjustments credibly. Your credentials make this page more trustworthy than generic sites.

4.7.1 Inputs

- Pre-pregnancy weight (kg)
- Height (cm)
- Age
- Trimester (1 / 2 / 3)
- Activity level
- Breastfeeding mode toggle (exclusive / partial / weaning)
- Number of fetuses (singleton / twins)

4.7.2 Formula (cite: IOM Pregnancy Weight Gain Guidelines 2009; Butte & King 2005)

Trimester / Stage	Additional Calories Above Pre-Pregnancy TDEE
1st Trimester	+0 kcal (no additional calories needed)
2nd Trimester	+340 kcal/day
3rd Trimester	+452 kcal/day
Twins — 2nd/3rd Trimester	+600 kcal/day (approximate, individualize with provider)
Exclusive Breastfeeding	+500 kcal/day (IOM 2009)
Partial Breastfeeding	+300 kcal/day
Weaning	+100 kcal/day

IMPORTANT

Display disclaimer prominently: 'These are evidence-based estimates. Caloric needs during pregnancy are highly individual. Always discuss nutrition with your OB, midwife, or registered dietitian.' This protects legally and reinforces credibility.

4.8 CALC-08: Recipe / Ingredient Nutrition Analyzer

WARNING

This is the most technically complex module. It requires browser-side HTTP calls to the USDA FoodData Central API. The API is free and does not require authentication for branded and foundation food data.

4.8.1 User Flow

1. User types or pastes a list of ingredients in a text area (one per line, with quantity, e.g. '200g chicken breast')
2. Click 'Analyze Recipe'
3. Each ingredient is parsed into quantity + unit + food name
4. App searches USDA FDC API for each food name, picks best match
5. Nutrient values are fetched for each match, scaled to the user's quantity
6. Results aggregated into a per-serving nutrition label (user sets serving count)
7. Label displayed in FDA Nutrition Facts panel format

4.8.2 USDA FoodData Central API

Endpoint	Purpose	Example
GET /foods/search?query={term}&api_key=DEMO_KEY	Search foods by name	/foods/search?query=chicken+breast
GET /food/{fdclid}?api_key=DEMO_KEY	Get full nutrient data for specific food	/food/171477
POST /foods	Batch fetch multiple FDC IDs in one request	Body: {fdclids: [171477, 174832]}

Base URL: <https://api.nal.usda.gov/fdc/v1>

API Key: Use 'DEMO_KEY' for development (1000 requests/hour/IP). Apply for a free key at api.data.gov for production (higher limits).

WARNING

No API key is stored in the codebase. 'DEMO_KEY' is a public string from USDA. Production key goes in a Cloudflare Pages environment variable (never committed to git).

4.8.3 Ingredient Parser (src/js/calculators/recipe-analyzer.js)

Each input line must be parsed with a regex that extracts:

- **quantity**: numeric value (integer or decimal)
- **unit**: g, kg, oz, lb, ml, l, cup, tbsp, tsp, piece, slice, medium, large, small
- **food**: remaining text after quantity and unit

```
// Regex pattern for ingredient parsing
const INGREDIENT_RE =
  /^(\d+(?:\.\d+)?(?:\s|\s+)?\s*
    (g|kg|oz|lb|ml|l|cup|tbsp?|tsp?|piece|slice|medium|large|small)?\s+
    (.+)\s$/i;

// Unit → grams conversion map
const UNIT_TO_GRAMS = {
  g: 1, kg: 1000, oz: 28.3495, lb: 453.592,
  ml: 1, l: 1000,
  cup: 240, tbsp: 15, tsp: 5,
  piece: null, // handled by USDA serving size
```

```
};
```

4.8.4 Nutrition Label Output

Display nutrients in the following order, formatted to resemble an FDA Nutrition Facts label:

- Calories (kcal)
- Total Fat (g) → Saturated Fat, Trans Fat, Unsaturated Fat
- Cholesterol (mg)
- Sodium (mg)
- Total Carbohydrate (g) → Dietary Fiber, Total Sugars, Added Sugars
- Protein (g)
- Vitamin D (mcg), Calcium (mg), Iron (mg), Potassium (mg)

Include a 'Copy Label' button (clipboard API) and a 'Print Label' button that opens print.css styles.

4.9 CALC-09: Diabetic Carb Counter

IMPORTANT

Low competition, high-intent niche. Users managing Type 1 or Type 2 diabetes need precise carb counting to match insulin doses. Very few free tools are designed specifically for this population.

4.9.1 Inputs

- Total daily carb target (g) — from their provider; editable
- Number of meals per day (3, 4, 5, 6)
- Carb distribution preference: Even / Front-loaded (larger breakfast) / Snack-adjusted
- Meal-by-meal manual entry mode (enter carbs per item, running total)

4.9.2 Outputs

- Carb budget per meal and per snack
- Running total tracker (add foods during the day)
- Visual gauge: % of daily carb budget used
- Insulin-to-carb ratio helper: if user inputs their I:C ratio, shows units of insulin needed per meal

IMPORTANT

Prominently state: 'Insulin dosing should always be managed with your diabetes care team. This calculator is a reference tool only, not a substitute for medical advice.' Add this disclaimer before the calculator, not just at the bottom.

NOTE

Cite: American Diabetes Association Standards of Medical Care in Diabetes 2024 (updated annually — verify citation year at publish time).

4.10 CALC-10: Kidney Diet Calculator (CKD / Dialysis)

IMPORTANT

Exceptional E-E-A-T opportunity. CKD diet management is extremely complex; no mainstream calculator handles it. Patients on dialysis must restrict potassium, phosphorus, and sometimes protein and fluid simultaneously. Your credentials matter immensely here.

4.10.1 Inputs

- CKD Stage (1–5, or Dialysis)
- Dialysis type (none / hemodialysis / peritoneal dialysis)
- Weight (kg)
- Urine output per day (yes/no; if no → add fluid restriction)

4.10.2 Output Targets (cite: KDIGO 2024, KDOQI Guidelines)

Nutrient	CKD 3–5 (Non-Dialysis)	Hemodialysis	Peritoneal Dialysis
Potassium	2,000–3,000 mg/day	2,000–3,000 mg/day	3,000–4,000 mg/day
Phosphorus	800–1,000 mg/day	800–1,000 mg/day	800–1,000 mg/day
Sodium	< 2,000 mg/day	< 2,000 mg/day	< 2,000 mg/day
Protein	0.6–0.8 g/kg/day	1.0–1.2 g/kg/day	1.2–1.5 g/kg/day
Fluid	Urine output + 500 ml	1,000–1,500 ml/day	Urine + PD volume

Output a daily limits dashboard with traffic-light visual indicators (green = within range, amber = approaching limit, red = over limit) that updates as the user logs food throughout the day.

IMPORTANT

Always recommend the user work with a renal dietitian (Registered Dietitian, RD). Display dietitian-finder link: <https://www.eatright.org/find-a-nutrition-expert>

5. Frontend Architecture & UI/UX

5.1 Design Tokens (CSS Custom Properties)

```
:root {  
  /* Colors */  
  --color-primary: #1A56A0; /* CTAs, links */  
  --color-primary-lt: #D6E4F7; /* Input focus rings, highlights */  
  --color-success: #1E6B3C; /* Valid inputs, success states */  
  --color-warning: #B45309; /* Warnings, amber states */  
  --color-danger: #9B1C1C; /* Errors, red states */  
  --color-text: #1A202C; /* Body text */  
  --color-text-muted: #64748B; /* Secondary text */  
  --color-bg: #FFFFFF;  
  --color-bg-alt: #F8FAFC; /* Alternating rows, card bg */  
  --color-border: #E2E8F0;  
  
  /* Typography */  
  --font-sans: 'Inter', system-ui, -apple-system, sans-serif;  
  --font-mono: 'JetBrains Mono', 'Fira Code', monospace;  
  --text-base: 1rem; /* 16px */  
  --text-sm: 0.875rem;  
  --text-lg: 1.125rem;  
  --text-xl: 1.25rem;  
  
  /* Spacing (8px base grid) */  
  --space-1: 0.25rem; --space-2: 0.5rem;  
  --space-3: 0.75rem; --space-4: 1rem;  
  --space-6: 1.5rem; --space-8: 2rem;  
  --space-12: 3rem; --space-16: 4rem;  
  
  /* Layout */  
  --max-width: 900px;  
  --sidebar-width: 280px;  
  --border-radius: 8px;  
  --shadow-sm: 0 1px 3px rgba(0,0,0,0.08);  
  --shadow-md: 0 4px 12px rgba(0,0,0,0.12);  
}
```

5.2 Page Layout (Calculator Pages)

Zone	Content	Width
Header	Site name, nav links (Home, All Calculators, About Author)	100%
Hero	H1 with calculator name, one-sentence description, trust badge (author name + credential)	100%

Zone	Content	Width
Calculator Card	Inputs, Calculate button, Results panel (all in one visible card)	Max 640px centered
Article	Full supporting article (how the formula works, when to use it, limitations)	Max 860px
Author Bio	Photo, name, credentials, 'Not Medical Advice' disclaimer	Full width, highlighted box
References	Numbered, linked to source DOIs where available	Below bio
Footer	Copyright, privacy policy link, disclaimer summary	100%

5.3 Calculator Card UX Requirements

- All inputs visible at once — no multi-step wizard (reduces friction and abandonment)
- Imperial/metric toggle persistent across the session (localStorage)
- Real-time validation: red border + message on blur if input is out of range; never on keystroke
- Calculate button disabled until all required fields are valid
- Results appear below inputs in the same card (no page reload, no scroll jump)
- Results include a 'Share Result' button (copies URL with query params encoding the inputs)
- Print button: triggers `window.print()` with `print.css` styles that hide nav/footer

5.4 Result Shareable URLs

Each calculator encodes inputs as URL query parameters so users can bookmark or share their results. Example:

```
/calculators/tdee/?age=30&sex=female&weight=65&height=165&activity=moderate&formula=mifflin
```

On page load, the calculator JS reads these params via `URLSearchParams`, pre-fills the form, and immediately calculates. This enables:

- Bookmarkable results
- Users sharing their specific scenario
- Internal linking between calculators (e.g., TDEE result → pre-fill TDEE into Macro calculator)

6. SEO Strategy & Content Architecture

6.1 Keyword Targeting by Calculator

CALC ID	Primary Target (Long-tail)	Secondary / Related	Est. Monthly Volume	Competition
CALC-01	tdee calculator mifflin st jeor	bmr calculator with activity level	8,000–15,000	High — target secondary first
CALC-02	macro calculator for cutting	macro ratio calculator muscle gain	5,000–12,000	Medium
CALC-03	navy body fat calculator	body fat percentage calculator circumference	2,000–5,000	Medium-Low
CALC-04	protein intake calculator by weight	how much protein per day kg	4,000–8,000	Medium
CALC-05	daily fiber intake calculator age	fiber needs per day women men	500–2,000	Low ✓
CALC-06	water intake calculator kg activity	daily water needs calculator	3,000–6,000	Medium
CALC-07	calorie calculator during pregnancy trimester	extra calories needed pregnant	1,000–3,000	Very Low ✓✓
CALC-08	recipe nutrition calculator paste ingredients	ingredient nutrition label calculator free	2,000–5,000	Low-Medium ✓
CALC-09	carb counting calculator diabetes	carbohydrate goal per meal type 2 diabetes	500–1,500	Very Low ✓✓
CALC-10	kidney diet potassium phosphorus calculator	ckd diet calculator stage 3	200–800	Extremely Low ✓✓✓

TIP

Start publishing CALC-07 (pregnancy), CALC-09 (diabetes), CALC-10 (kidney) first. These have almost no real competition from credentialed sources. You can rank within 30–60 days for these terms even with a new domain.

6.2 On-Page SEO Requirements (Every Calculator Page)

Element	Requirement	Example
<title>	Primary keyword + brand. Under 60 chars.	TDEE Calculator (Mifflin-St Jeor) NutriCalc Pro
<meta description>	150–160 chars. Include keyword + action + differentiator.	Calculate your TDEE using the Mifflin-St Jeor formula. Multiple activity levels, Katch-McArdle option. Written by [Name], MS RD.
H1	Exact match or close variant of primary keyword	TDEE Calculator: Mifflin-St Jeor & Katch-McArdle
H2s in article	Answer related questions ('What is TDEE?', 'How accurate is this formula?')	—
Canonical URL	<link rel='canonical' href='https://...> on every page	—
Open Graph	og:title, og:description, og:image (1200×630 PNG per calculator)	—
Alt text on images	Descriptive, keyword-relevant	TDEE formula diagram showing Mifflin-St Jeor equation
Internal links	Each calculator article links to 2–3 related calculators contextually in the article body	—
Word count	Supporting article: 800–1500 words. Original. Not AI-published directly.	—

6.3 Structured Data (JSON-LD Schema)

Every page must include the following schema types in a <script type='application/ld+json'> block in the <head>:

- **WebApplication**: for the calculator widget itself — name, url, applicationCategory: 'HealthApplication'
- **Article / MedicalWebPage**: for the supporting article — author, datePublished, dateModified, citation array
- **Person**: for the author — name, credentials, sameAs (link to LinkedIn, institution profile)
- **BreadcrumbList**: Home > Calculators > [Calculator Name]
- **FAQPage**: 3–5 common questions answered in the article, marked up for SERP rich results

```
{
  "@context": "https://schema.org",
```



```
"@type": "WebApplication",
"name": "TDEE Calculator — Mifflin-St Jeor",
"url": "https://nutricacpro.com/calculators/tdee/",
"applicationCategory": "HealthApplication",
"operatingSystem": "All",
"offers": { "@type": "Offer", "price": "0" },
"author": {
  "@type": "Person",
  "name": "[Your Full Name]",
  "honorificSuffix": "MS, RD",
  "sameAs": "https://www.linkedin.com/in/yourprofile"
}
```

6.4 Author Page & E-E-A-T Strategy

Google's E-E-A-T (Experience, Expertise, Authoritativeness, Trustworthiness) signals are the primary lever for medical/health content to rank in 2024–2026. This site's primary competitive advantage is a real, named, credentialed author.

6.4.1 Author Page Requirements (/about/)

- Full name, professional headshot (real photo, not AI-generated)
- Credentials listed precisely: degrees (field, institution, year), certifications (body, year), professional memberships
- Brief professional bio (200–300 words): training, clinical/research experience, why you built this site
- Link to LinkedIn profile, ResearchGate, institutional profile, or published work if any
- Statement of editorial independence ('No sponsored content. No advertiser relationships.')
- Review policy: 'All calculator formulas are reviewed against the cited literature annually'

6.4.2 Author Bio Block (appears on every calculator page)

The author bio component (src/_includes/components/author-bio.njk) must render:

- Circular headshot (64×64px on mobile, 80×80px desktop)
- Name, credential suffix, 'Reviewed [Month Year]'
- Two-sentence blurb from site.json (editable globally)
- Link to full /about/ page
- Immediately followed by disclaimer.njk

7. Performance Requirements & Core Web Vitals

7.1 Performance Targets

Metric	Target	Measurement Tool
LCP (Largest Contentful Paint)	< 1.5 seconds	PageSpeed Insights, CrUX
FID / INP (Interaction to Next Paint)	< 100ms	PageSpeed Insights
CLS (Cumulative Layout Shift)	< 0.05	PageSpeed Insights
TTFB (Time to First Byte)	< 100ms	WebPageTest (Cloudflare edge serves static files)
PageSpeed Score (Mobile)	> 95	PageSpeed Insights
PageSpeed Score (Desktop)	> 99	PageSpeed Insights
Total Page Weight (HTML+CSS+JS)	< 150 KB uncompressed	Chrome DevTools Network tab

7.2 Achieving Targets

- **No JavaScript frameworks:** Vanilla JS only. No React, no Vue, no jQuery. A 30-line vanilla JS calculator is ~2KB. React is 130KB+ before your code.
- **CSS: one file, no unused rules:** No utility framework. Write only the CSS you use. Use CSS Grid and Flexbox — no JS-driven layout.
- **Fonts:** Load Inter via Google Fonts with display=swap and preconnect. Subset to Latin only (saves 60% vs full font). Or use system-ui fallback with no external font request at all.
- **Images:** WebP format. Serve via <picture> with JPEG fallback. Lazy-load below-the-fold images. Author headshot: 80×80px actual dimensions, not CSS-scaled down from 800px.
- **Inline critical CSS:** < 14KB of above-the-fold styles inline in <head>. Load rest asynchronously via media='print' trick.
- **Cloudflare caching:** Static assets (CSS, JS, images) get Cache-Control: public, max-age=31536000 with content-hash filenames. HTML pages: cache for 1 hour at edge, revalidate.
- **Minification:** HTML minified by 11ty transform. JS and CSS minified with esbuild (JS) and lightningcss (CSS) in the build pipeline.

8. Deployment, CI/CD & Infrastructure

8.1 Infrastructure Overview

Component	Service	Cost
Domain	Namecheap / Cloudflare Registrar	~\$10/year
DNS	Cloudflare (free)	\$0
Hosting / CDN	Cloudflare Pages (free tier: unlimited bandwidth, 500 builds/month)	\$0
SSL Certificate	Cloudflare (automatic, included)	\$0
Analytics	Cloudflare Web Analytics (privacy-first)	\$0
USDA API	api.data.gov (free, 1000 req/hour with free key)	\$0
Source Control	GitHub (free)	\$0
Build System	GitHub Actions (2000 min/month free)	\$0
Total Monthly Cost	—	\$0 (until domain renewal)

8.2 GitHub Actions Pipeline (.github/workflows/deploy.yml)

```
name: Build and Deploy
on:
  push:
    branches: [main]

jobs:
  build-and-deploy:
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v4

      - name: Setup Node.js
        uses: actions/setup-node@v4
        with:
          node-version: '20'
          cache: 'npm'

      - name: Install dependencies
        run: npm ci

      - name: Build site
        run: npm run build
```

```
env:
  USDA_API_KEY: ${ secrets.USDA_API_KEY }

- name: Deploy to Cloudflare Pages
  uses: cloudflare/pages-action@v1
  with:
    apiToken: ${ secrets.CF_API_TOKEN }
    accountId: ${ secrets.CF_ACCOUNT_ID }
    projectName: nutri-calc-pro
    directory: _site
    githubToken: ${ secrets.GITHUB_TOKEN }
```

8.3 Environment Variables

Variable	Where Set	Value
USDA_API_KEY	Cloudflare Pages → Settings → Environment Variables	Your api.data.gov key (or 'DEMO_KEY' for dev)
CF_API_TOKEN	GitHub repository → Settings → Secrets	Cloudflare API token with Pages:Edit permission
CF_ACCOUNT_ID	GitHub repository → Settings → Secrets	Your Cloudflare account ID

IMPORTANT

NEVER commit API keys to the repository. The USDA API key is injected at build time via GitHub Actions secrets. The browser-side calls use it only if you build it into the JS at build time — OR use 'DEMO_KEY' in the browser-facing JS (acceptable for USDA public endpoints).

9. Security, Privacy & Legal Requirements

9.1 HTTP Security Headers

Configure via a `_headers` file at the root of the `_site` output (Cloudflare Pages reads this file):

Header	Value	Purpose
Content-Security-Policy	default-src 'self'; script-src 'self'; style-src 'self' fonts.googleapis.com; font-src fonts.gstatic.com; connect-src api.nal.usda.gov	Prevents XSS; allows only USDA API calls
X-Frame-Options	DENY	Prevents clickjacking
X-Content-Type-Options	nosniff	Prevents MIME sniffing
Referrer-Policy	strict-origin-when-cross-origin	Limits referrer info sent to external sites
Permissions-Policy	geolocation=(), microphone=(), camera=()	Disables browser APIs the site doesn't use
Strict-Transport-Security	max-age=63072000; includeSubDomains; preload	Forces HTTPS; enables HSTS preloading

9.2 Privacy Compliance

- No cookies set by the site (Cloudflare Web Analytics is cookieless)
- No user data is transmitted to any server by calculators (except recipe analyzer → USDA API, which receives only food names — no user identity)
- No third-party scripts (no Google Tag Manager, no Meta Pixel, no Hotjar)
- Privacy Policy page required at `/privacy/` — disclose Cloudflare's data processing and the USDA API query data
- GDPR / CCPA: Because no PII is collected and no cookies are set, consent banners are not legally required — but include a brief notice in the footer

9.3 Medical / Legal Disclaimers

IMPORTANT

All calculator pages **MUST** display this disclaimer. It must be visible without scrolling on desktop (place it in or immediately below the calculator card, not buried at the bottom). Failure to do this creates legal liability.

Required disclaimer text (exact wording, customize as needed for your jurisdiction):

Educational Purposes Only — Not Medical Advice

The results provided by this calculator are estimates based on population-level research formulas. They are intended for educational and informational purposes only. Individual needs vary based on health status, medications, metabolic conditions, and other factors not captured by these formulas.

This tool is not a substitute for personalized advice from a licensed healthcare provider, registered dietitian, or physician. Always consult a qualified professional before making significant changes to your diet, exercise routine, or health management plan.

10. Development Phases & Task Breakdown

Phase 1 — Foundation (Week 1–2)

Goal: Working static site, deployed, with 2 calculators live.

Task	Priority	Estimated Hours
Initialize 11ty project, install dependencies, configure .eleventy.js	P0	2h
Create base.njk and calculator.njk layouts with full HTML structure	P0	4h
Build CSS: design tokens, reset, typography, calculator card, responsive grid	P0	6h
Build src/js/core/units.js (all unit conversions) with unit tests	P0	3h
Build src/js/core/validate.js (all input validation rules)	P0	2h
Build CALC-01: TDEE/BMR (both formulas) with shareable URL support	P0	4h
Build CALC-07: Pregnancy Calorie Calculator	P0	3h
Write author bio content, about page	P0	2h
Configure GitHub repo, GitHub Actions workflow	P0	2h
Deploy to Cloudflare Pages, connect custom domain, verify SSL	P0	2h
Write supporting articles for CALC-01 and CALC-07 (original content)	P0	8h
Submit sitemap to Google Search Console	P0	0.5h

Phase 2 — Core Calculators (Week 3–5)

Goal: All client-side calculators complete, SEO-optimized, articles written.

Task	Priority	Estimated Hours
Build CALC-02: Macro Calculator (all modes, custom ratios, real-time validation)	P1	4h
Build CALC-03: Navy Body Fat Calculator	P1	3h
Build CALC-04: Protein Intake Calculator	P1	3h

Task	Priority	Estimated Hours
Build CALC-05: Fiber Intake Calculator	P1	2h
Build CALC-06: Water Intake Calculator with schedule widget	P1	4h
Build CALC-09: Diabetic Carb Counter (meal tracker, I:C ratio helper)	P1	5h
Build CALC-10: Kidney Diet Calculator (traffic-light dashboard)	P1	6h
Write supporting articles for all 7 new calculators	P1	20h
Implement JSON-LD schema on all pages	P1	3h
Add Open Graph meta tags, social share images	P1	2h
Cross-link all calculators contextually in article bodies	P1	2h
Accessibility audit: all inputs ARIA-labeled, keyboard navigable	P1	4h
Performance audit: all pages > 95 PageSpeed Mobile	P1	3h

Phase 3 — Recipe Analyzer (Week 6–8)

Goal: Recipe analyzer fully functional, ingredient parser robust, nutrition label accurate.

Task	Priority	Estimated Hours
Build ingredient parser with regex + unit conversion	P2	4h
Integrate USDA FoodData Central search endpoint	P2	3h
Build food selection UI (show top 3 matches, let user pick)	P2	4h
Build nutrient aggregator + serving scaler	P2	3h
Build FDA-style Nutrition Facts panel renderer	P2	5h
Implement Copy Label (clipboard API) and Print functions	P2	2h
Error handling: rate limits, no results found, parse failures	P2	2h

Task	Priority	Estimated Hours
Write supporting article for recipe analyzer	P2	3h
Load and stress test: 20-ingredient recipes, bulk analysis	P2	2h

Phase 4 — Polish & SEO Monitoring (Month 2–3)

Goal: Monitor rankings, refine content, fix gaps, add FAQPage schema, build backlinks.

- Check Google Search Console weekly for indexing issues, CTR, and ranking movement
- Add or expand FAQ sections on pages that aren't ranking (FAQ schema helps with featured snippets)
- Write 1–2 supporting blog articles per month: 'What is TDEE and why does it matter?', 'How to count carbs for diabetes: a step-by-step guide'
- Submit to relevant directories: dietitian resource lists, diabetes education sites, pregnancy apps (where they link to external tools)
- Set up Cloudflare Analytics dashboard review cadence (weekly)

11. Testing Requirements

11.1 Calculator Formula Testing

Every calculator must have a test file in /tests/ that verifies formula accuracy against known reference values from the published literature.

Calculator	Test Case (Input → Expected Output, ±1 kcal/g rounding)
TDEE/BMR	30yo female, 65kg, 165cm, moderate activity → Mifflin BMR: 1,447 kcal, TDEE: 2,243 kcal
TDEE/BMR	40yo male, 85kg, 180cm, sedentary → Mifflin BMR: 1,877 kcal, TDEE: 2,252 kcal
Katch-McArdle	70kg, 20% BF (56kg LBM) → BMR: 1,579 kcal
Macros (Cut)	TDEE 2,200 kcal, cut mode → 1,760 kcal, Carbs: 176g, Protein: 154g, Fat: 49g
Navy BF (Male)	Height 180cm, waist 85cm, neck 38cm → ~18.5% BF
Water	70kg, moderate activity, temperate → ~2,800 ml/day
Pregnancy	65kg, 165cm, 25yo, 2nd trimester, moderate → ~2,583 kcal

11.2 Input Validation Testing

- All fields: empty → required error message
- Age: 0 → error; 14 → error; 15 → valid; 100 → valid; 101 → error
- Body fat: -1 → error; 0 → error; 3 → valid; 60 → valid; 61 → error
- Weight: 0 → error; 29 → error; 30 → valid; 300 → valid; 301 → error
- Macro custom ratios: 40/30/30 → valid (sum=100); 40/30/29 → error (sum=99)

11.3 Accessibility Testing

- All form inputs have associated <label> elements (not just placeholder text)
- All results announced to screen readers via aria-live='polite' region
- Tab order is logical: inputs → calculate button → results
- Color is not the only indicator of state (errors use icon + text + border color)
- Test with: NVDA (Windows) + Chrome; VoiceOver (macOS) + Safari
- axe DevTools browser extension: zero violations on every page

11.4 Cross-Browser Testing

Browser + OS	Must Pass
Chrome 120+ / Windows 11	Full functionality
Firefox 120+ / Windows 11	Full functionality
Safari 17 / macOS Sonoma	Full functionality (test CSS Grid, URLSearchParams)
Chrome / Android 13 (Pixel)	Touch targets $\geq 48\text{px}$, no horizontal scroll
Safari / iOS 17 (iPhone 14)	All inputs usable with iOS keyboard, no zoom on focus

12. Scientific References

All calculator implementations must cite these sources in their supporting articles. Link to DOI where available.

#	Citation	Used In
1	Mifflin MD, et al. (1990). A new predictive equation for resting energy expenditure in healthy individuals. <i>American Journal of Clinical Nutrition</i> , 51(2), 241–247. DOI: 10.1093/ajcn/51.2.241	CALC-01
2	McArdle WD, Katch FI, Katch VL. (1996). <i>Exercise Physiology: Energy, Nutrition, and Human Performance</i> . 4th ed. Lippincott Williams & Wilkins.	CALC-01
3	Hodgdon JA, Beckett MB. (1984). Prediction of percent body fat for U.S. Navy men and women from body circumferences and height. <i>Naval Health Research Center Report No. 84–29</i> .	CALC-03
4	Morton RW, et al. (2018). A systematic review, meta-analysis and meta-regression of the effect of protein supplementation on resistance training-induced gains in muscle mass and strength. <i>British Journal of Sports Medicine</i> , 52(6), 376–384.	CALC-04
5	Jäger R, et al. (2017). International Society of Sports Nutrition Position Stand: Protein and exercise. <i>Journal of the International Society of Sports Nutrition</i> , 14, 20.	CALC-04
6	Institute of Medicine. (2005). <i>Dietary Reference Intakes for Energy, Carbohydrate, Fiber, Fat, Fatty Acids, Cholesterol, Protein, and Amino Acids</i> . National Academies Press.	CALC-05, CALC-06
7	EFSA Panel on Dietetic Products, Nutrition, and Allergies. (2010). Scientific opinion on dietary reference values for water. <i>EFSA Journal</i> , 8(3), 1459.	CALC-06
8	Institute of Medicine. (2009). <i>Weight Gain During Pregnancy: Reexamining the Guidelines</i> . National Academies Press. DOI: 10.17226/12584	CALC-07
9	Butte NF, King JC. (2005). Energy requirements during pregnancy and lactation. <i>Public Health Nutrition</i> , 8(7A), 1010–1027.	CALC-07
10	American Diabetes Association. (2024). Standards of Medical Care in Diabetes. <i>Diabetes Care</i> , 47(Supplement 1).	CALC-09
11	Kidney Disease: Improving Global Outcomes (KDIGO). (2024). KDIGO Clinical Practice Guideline for Chronic Kidney Disease Evaluation and Management.	CALC-10
12	Kidney Disease Outcomes Quality Initiative (KDOQI). (2020). KDOQI Clinical Practice Guideline for Nutrition in CKD. <i>American Journal of Kidney Diseases</i> , 76(3), S1–S107.	CALC-10

13. Appendix

A. Package.json Scripts

```
{
  "scripts": {
    "dev": "eleventy --serve --incremental",
    "build": "NODE_ENV=production eleventy",
    "test": "node --test tests/**/*.test.js",
    "lint": "eslint src/js/**/*.js",
    "ally": "axe-cli http://localhost:8080 --exit"
  },
  "dependencies": {
    "@11ty/eleventy": "^3.0.0",
    "eleventy-plugin-sitemap": "^2.1.0",
    "markdown-it": "^14.0.0",
    "markdown-it-anchor": "^9.0.0"
  },
  "devDependencies": {
    "html-minifier-terser": "^7.2.0",
    "esbuild": "^0.23.0",
    "lightningcss": "^1.25.0",
    "@axe-core/cli": "^4.9.0"
  }
}
```

B. File Naming Conventions

Entity	Convention	Example
JS calculator modules	kebab-case, matches calc slug	src/js/calculators/kidney-diet.js
Nunjucks page files	kebab-case, becomes URL segment	src/calculators/kidney-diet.njk → /calculators/kidney-diet/
CSS files	descriptive, single-purpose	src/css/calculator.css
Test files	*.test.js suffix	tests/tdee.test.js
Article markdown	matches calculator slug	src/articles/tdee.md
Images	descriptive-slug.webp	src/images/pregnancy-calorie-chart.webp

C. USDA FoodData Central Nutrient IDs

When parsing USDA API responses, use these nutrient IDs to extract values reliably across food types:

Nutrient	USDA Nutrient ID
Energy (kcal)	1008
Protein (g)	1003
Total Fat (g)	1004
Saturated Fat (g)	1258
Trans Fat (g)	1257
Carbohydrates (g)	1005
Dietary Fiber (g)	1079
Total Sugars (g)	2000
Sodium (mg)	1093
Potassium (mg)	1092
Phosphorus (mg)	1091
Calcium (mg)	1087
Iron (mg)	1089
Vitamin D (mcg)	1114
Cholesterol (mg)	1253

D. Revision History

Version	Date	Author	Changes
1.0.0	June 16, 2026	[Your Name]	Initial SRS document